Question

There are the following five processes:

1. In a warm acetonitrile solution, sodium fluoride reacts with sulfur dichloride, and one of the products is a yellow-red liquid;

2. Under high temperature and pressure, Ni reacts with PF_3 to form a tetrahedral molecule;

3. Passing sufficient carbon dioxide into sodium metaborate produces borax crystals;

4. A small amount of ClF_3 added to a solution of hydrazine reacts violently;

5. Copper reacts with dinitrogen tetroxide in nitromethane, and the product contains Cu 22.73%.

Based on the chemical equations (with the simplest integer coefficients) of the above five processes, which of the following statements is incorrect

A. Process 1 The sum of the coefficients in the chemical reaction equation is 13.

B. Process 2: The sum of the coefficients in the chemical reaction equation is 6.

C. The sum of the stoichiometric coefficients of the reactants in process 3 is 17.

D. The sum of the coefficients of the reactants in process 4 is 6.

E. The product of Process 5 is an important biomolecule

F. The products of process 4 are all diatomic small molecules.

G. All of the above options are incorrect.

Answer

Correct Answer: F

Detailed Explanation

This is a chemistry equation problem, mainly examining the deduction of inorganic reaction products.

In process 1, the yellow-red liquid containing S and Cl is $\mathrm{S_2Cl_2}$, which can be used to determine the destination of S and Cl. Since NaF participates in the reaction, the only destination of Na is to combine with Cl, and the only destination of F is to combine with S, forming the common $\mathrm{SF_4}$.

CHECKPOINT 1 PTS

Products are $\mathrm{S_2Cl_2}$, NaCl, and $\mathrm{SF_4}$

Therefore, the balanced equation is $4\mathrm{NaF} + 3\mathrm{SCl_2} = \mathrm{S_2Cl_2} + 4\mathrm{NaCl} + \mathrm{SF_4}$, and the sum of the coefficients is 13, so option A is correct.

CHECKPOINT 0.5 PTS

The sum of the coefficients of the balanced equation in process 1 is 13

In process 2, the tetrahedral molecule is $\mathrm{Ni(PF_3)_4}$, and the balanced equation is $\mathrm{Ni} + 4\mathrm{PF_3} = \mathrm{Ni(PF_3)_4}$, and the sum of the coefficients is 6, so option B is correct.

CHECKPOINT 1 PTS

The tetrahedral molecule corresponds to four ligands, and the sum of the coefficients of the balanced equation is 6

In process 3, borax crystal is $Na_2B_4O_7 \cdot 10H_2O$. In addition, note that CO_2 is in sufficient quantity, and the formed Na_2CO_3 will further react with CO_2, and the product should be $NaHCO_3$.

CHECKPOINT 1 PTS

Borax is $Na_2B_4O_7 \cdot 10H_2O$, and the reaction product is $NaHCO_3$

Therefore, the balanced equation is $4NaBO_2 + 2CO_2 + 11H_2O = Na_2B_4O_7 \cdot 10H_2O + 2NaHCO_3$, and the sum of the reactant coefficients is 17, so option C is correct.

CHECKPOINT 0.5 PTS

The sum of the reactant coefficients of the balanced equation in process 2 is 17

In process 4, the question mentions "violent reaction", indicating that a violent redox reaction has occurred, and N_2 is produced. $Cl(III)$ has strong oxidizing properties and will be reduced to a valence of -1. It should be pointed out that since 0-valent Cl (i.e., Cl_2) also has strong oxidizing properties, it will continue to react with N_2H_4, and Cl_2 should not appear in the products.

CHECKPOINT 1 PTS

In the products of process 4, Cl has a valence of -1

The question indicates that the reaction environment is a hydrazine solution, and attention should be paid to the existence form of the products. The generated HCl and HF will combine with hydrazine to form $\mathrm{N_2H_5Cl}$ and $\mathrm{N_2H_5F}$.

CHECKPOINT 1 PTS

Reaction products should combine with hydrazine to form $\mathrm{N_2H_5Cl}$ and $\mathrm{N_2H_5F}$

Therefore, the balanced chemical equation is $\mathrm{ClF_3 + 5N_2H_4 = N_2 + N_2H_5Cl + 3N_2H_5F}$. The sum of the coefficients of the reactants is 6, so option D is correct; the products are not only diatomic molecules, so option F is incorrect.

CHECKPOINT 0.5 PTS

The sum of the reactant coefficients in process 4 is 6

In process 5, let's assume the product is $\mathrm{CuN_xO_y}$, solve the indeterminate equation $\frac{63.55}{63.55+14.01x+16y} = 0.2273$, where x and y are both integers. One set of solutions is $x = 4, y = 10$, i.e., $\mathrm{CuN_4O_{10}}$. Considering element conservation, 3 $\mathrm{N_2O_4}$ leaving the product $\mathrm{CuN_4O_{10}}$ still have two N and two O, i.e., two NO molecules.

CHECKPOINT 1 PTS

The reaction product of process 5 is CuN_4O_{10}

Therefore, the balanced chemical equation is $Cu + 3N_2O_4 = CuN_4O_{10} + 2NO$, in fact, it is $Cu(NO_3)_2 \cdot N_2O_4$. Since NO is an important biological molecule, option E is correct.

CHECKPOINT 1 PTS

Another product of process 5 is NO

CHECKPOINT 1 PTS

Option D is incorrect, A, B, C, E, and F are all correct